

ASSIGNMENTS**Class-10**

- (a) If a 7 by 9 matrix has rank 5, what are the dimensions of the four subspaces?
What is the sum of all four dimensions?
- (b) If a 3 by 4 matrix has rank 3, what are its column space and left nullspace?

c)

Without elimination, find dimensions and bases for the four subspaces for

$$A = \begin{bmatrix} 0 & 3 & 3 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 \\ 4 \\ 5 \end{bmatrix}.$$